

Special Notice DARPA-SN-26-67

Defense Advanced Research Projects Agency (DARPA) / Biological Technologies Office (BTO) Research Opportunity: DNA-Protein Epigenetic Chemistry Engineering (D-PECHE)

1. INTRODUCTION

This Special Notice describes DNA-Protein Epigenetic Chemistry Engineering (D-PECHE), a research thrust that is focused on the rapid evaluation of enzymatic pathways for generating epigenetic modifications. This Special Notice is released under the current DARPA BTO Office Wide Broad Agency Announcement (BAA), HR001126S0003, entitled “Biological Technologies,” which can be found at the link below:

<https://sam.gov/opp/8d403582edfd409795560247e8d229b7/view>

This Special Notice is soliciting proposal abstracts to HR001126S0003. Proposers are required to submit a proposal abstract (2 pages maximum). The requirements for the abstract submission, evaluation, and award of any resulting contracts will be subject to the terms described in HR001126S0003.

- Proposal Abstract Due Date: May 27th, 2026
- Oral Proposal Packages due approximately 4 weeks after invitation

Late abstract submissions may not be evaluated. **(NOTE: The above-listed deadlines differ from those listed in HR001126S0003)**

The purpose of this announcement is (1) to draw the attention of the scientific community to the D-PECHE study topics and (2) to initiate dialogue and teaming that will lead to abstract/OPP submissions to address the D-PECHE study topics.

2. TOPIC DESCRIPTION

DARPA is soliciting innovative proposals to demonstrate proof-of-concept for novel mechanisms for writing and erasing epigenetic base modifications in DNA. Mechanisms capable of modifying the epigenome offer new opportunities to control DNA-protein interactions and to study the functions and applications of rare or novel epigenetic modifications. This research may result in demonstrations of novel technologies to modify nucleic acids *in vitro*.

Overview – D-PECHE:

The epigenome serves as a critical regulatory layer governing gene expression, cellular identity, and organismal response to environmental stimuli. Epigenetic modifications such as DNA methylation, histone post-translational modification, and chromatin remodeling can regulate gene expression and DNA-protein interactions without altering the underlying genetic sequence. Unlike permanent genome editing approaches, epigenetic modifications are potentially reversible and offer a compelling avenue for dynamic, context-sensitive biological control. The ability to precisely modulate DNA-protein interactions has profound implications for fields ranging from regenerative medicine and cognitive performance

improvements to countermeasures against biological threats and environmental stressors. D-PECHE hypothesizes that novel DNA modification mechanisms that alter the biochemical properties of DNA are required to unlock new applications of epigenetics.

Epigenetic editor platforms are constrained by the number of effective writers and erasers for targeted modification of DNA bases. Many epigenetic editing platforms rely on DNA methylation and demethylation to regulate gene expression. Additionally, base editors use engineered deaminases to trigger specific base substitutions during DNA replication. These editing platforms only explore a small number of potential base modifications. Programmable writer and eraser proteins do not exist for the majority of naturally occurring base modifications, limiting translational applications of these base modifications. Writing and erasing new-to-nature base modifications will require new technologies capable of catalyzing new chemical modifications to DNA *in vitro*.

The introduction of novel base modifications into genomes raises fundamental questions about cellular compatibility that do not arise with well-studied modifications such as 5-methylcytosine (5mC). Cells have base excision repair (BER) and DNA damage response (DDR) pathways that continuously survey the genome for DNA adducts and mismatches to initiate repair or apoptotic responses. Novel base modifications that are not recognized as canonical epigenetic modifications may be misidentified as DNA damage by these surveillance systems, triggering unintended repair reactions that remove the modification, introduce mutations, or activate cell death pathways. Additionally, installation of novel base modifications will require the availability of appropriate metabolic precursors and cofactors. New epigenetic editing platforms may need to develop mitigation strategies for these challenges, which may affect their potential applications.

Erasure of novel base modifications requires both removal of the chemical modification as well as restoration of the correct unmodified base. Current approaches to removing epigenetic modifications include TET enzymes, which initiate demethylation of 5mC into an unmethylated cytosine. Such approaches are modification specific and are unlikely to be effective erasers for other proteins. Future applications of epigenetic editors will require new approaches to erasing epigenetic modifications and restoring unmodified bases. Such approaches could include modification agnostic eraser enzymes capable of removing multiple different modifications, or targeted BER activation for complete removal and replacement of modified bases.

D-PECHE performers must leverage expertise in nucleic acid chemistry, protein engineering, and synthetic biology to overcome challenges in writing and erasing epigenetic modifications. If successful, this will derisk a broader concept of epigenetic editing platforms and their potential applications.

Objective:

DARPA BTO seeks to de-risk development of new epigenetic editing mechanisms in a 4-month period of performance. An evaluation of technical progress will occur every month, and successful fulfillment of milestones and intermediate metrics will be required for the continuation of work.

Scope of Work:

The goal of the effort is to develop novel methods of modifying DNA bases in living cells and validate the presence of new base modifications catalyzed *in vitro*. For this effort, DARPA is interested in base modifications without prior *in vitro* demonstrations. Work under this effort will require performers to identify a specific DNA modification to write or erase in a cell line of the performer's choice. DARPA has no preference for *in vitro* demonstrations in eukaryotes, bacteria, or archaea; however, performers working on epigenetic writers must show that their targeted base modifications do not natively occur in their working cell line.

DNA-modifying mechanisms must 1) place the target base in the catalytic active site, 2) recruit cofactors and other metabolic precursors, and 3) restore the correct base orientation after modification. Considerations for modification selection may include (but are not limited to): known enzymatic pathways that catalyze the selected base modification; mechanistic relevance for regulating DNA-protein interactions; availability of putative co-factors for base modification; and availability of methods to detect base modifications. Initial screens of target modifications and enzyme candidates may be performed as a risk mitigation strategy, but the selection of targets and enzymes must be technically justified. Abstracts and proposals that use non-enzymatic approaches, such as chemical conjugation, to modify bases are considered in scope.

Performers may choose to work on epigenetic writing mechanisms, erasing mechanisms, or both. Performers who choose to develop epigenetic erasing mechanisms do not need to implement novel modifications to validate eraser function; however, they must demonstrate the novelty of their proposed erasing mechanism and describe their approach to validating eraser function. Potential validation approaches may validate efficacy against suitable *in vitro* targets or demonstrate generalizability of eraser function to multiple base modifications. Approaches could include but are not limited to the incorporation of base analogues or DNA adducts into living cells to generate eraser targets for validation.

At the conclusion of the period of performance, the D-PECHE research opportunity aims to demonstrate the following:

- 1) An ability to design novel mechanisms for writing or erasing targeted base modifications,
- 2) An ability to use novel mechanisms to modify the epigenome of living cells, and
- 3) An ability to detect the presence of targeted base modifications.

Out of Scope:

Abstracts and OPPs will be considered out of scope if they 1) only propose to identify DNA-modifying enzymes without subsequent *in vitro* validation in different cell lines or organisms, 2) propose to develop new cell-delivery or targeting mechanisms for epigenetic writers and erasers, 3) propose improvements to established epigenetic editing platforms without altering the substrate specificity or targeted modification of epigenetic editors, or 4) use sequences that would be export controlled or restricted for biosafety reasons. Additionally, this research opportunity will not support development of DNA methylases or demethylases. All human subjects and animal subjects research is explicitly out of scope for D-PECHE.

Anticipated Structure:

The D-PECHE study duration is 4 months. DARPA intends to make multiple awards under this research opportunity. The period of performance (PoP) includes establishing the team, technologies, and experimental pipeline to achieve milestones outlined in the section below.

Proposers shall develop a milestone-driven work plan, incorporating metrics and deliverables into the 4-month PoP. Proposers shall clearly identify and justify metrics for evaluating the feasibility of their approach. Abstracts and proposals should include measurement methodologies for detecting modified DNA to achieve the D-PECHE study goals of developing novel methods of modifying DNA bases in living cells and validating the presence of new base modifications catalyzed *in vitro*.

The D-PECHE study will consist of 2 main tasks:

Task 1: Demonstrate the ability to design DNA-modifying mechanisms to write and/or erase DNA base modifications. Identify target modification, catalyst candidates, and substrates and cofactors required for base modification. Identify any changes to protein sequences or domains needed for functionality within living cells.

Task 2: Detect DNA modifications *in vitro*. Demonstrate biocompatibility of targeted modification in host cell line. Develop assays to detect base modifications. Establish approaches to detect novel modifications in cells.

Table 1: D-PECHE Task Overview

Task	Potential Approaches
Task 1: Design of new DNA modification mechanisms	Bioprospecting for DNA-modifying enzymes, directed evolution of epigenetic editors, <i>de novo</i> design of DNA-modifying enzymes, discovery of catalysts with high DNA specificity
Task 2: Detection of DNA modifications <i>in vitro</i>	Mass spectroscopy of DNA extracts, enzymatic assays (i.e., ELISA) to target reactive modifications, engineering of reporter cell lines

Table 1: D-PECHE task overview. Please note that the table is not meant to be prescriptive, comprehensive, or exclusionary.

Deliverables:

All proposals must provide the following deliverables as part of their work plan:

- A kickoff meeting that provides a technical plan to accomplish the D-PECHE study goals;
- Monthly status reports that detail progress towards milestone
- A final report that provides a comprehensive overview of all progress towards the D-PECHE study goals and milestones and validates their research progress under this effort with supporting data, figures, and/or code.

Technical Guidance for Abstracts and OPPs:

As described in the BTO Office-wide BAA (HR001126S0003), respondents to this Special Notice must submit abstracts, and based on an evaluation of those abstracts, DARPA may invite respondents to submit an OPP. Abstracts in response to this Special Notice are limited to 2 pages, excluding the title page and references.

Abstracts and OPPs should:

- Clearly state whether they plan to demonstrate epigenetic writing mechanisms, erasing mechanisms, or both. Proposers that plan to develop writing mechanisms should describe the target DNA modification and how key properties of the modification inform their approach to detection of DNA modifications or validation of writing mechanism function. Proposers that plan to work on epigenetic erasers should describe the conceptual mechanisms for removal of epigenetic modifications as well as potential methods to validate eraser function.
- Describe their approach to mechanism development and the feasibility of development within the D-PECHE research opportunity timeline. Feasibility can be justified through references to prior art and/or computational, rational, or empirical methods for achieving novel epigenetic functions within living cells.
- Describe planned experiments for *in vitro* validation of epigenetic writers and erasers, including direct or indirect approaches to detect base modifications or eraser function.
- Describe the cell line to be used in development of epigenetic writers or erasers and any work to develop reporter, knockout, and/or engineered cell lines for *in vitro* validation.
- Provide a statement that the research contemplated in the project does not duplicate research being done under other DoD programs or Government entities.

3. SUBMISSION AND AWARD INFORMATION

This announcement **requires the use of the accelerated award** for selected proposals. Awards for such proposals would be made within 30 days of the selection notification. Proposers must agree to prescribed stipulations as part of their OPP submissions. Attachments G and L must be included in the OPP submission to receive this type of award. Proposers must request this accelerated award option and must submit through the Broad Agency Announcement Tool (BAAT). Attachment G with the selection for Other Transactions for Research (OT-R) will be applicable to this Special Notice.

DARPA anticipates that all abstracts and proposals received in response to this special notice will be unclassified. D-PECHE will be conducted as unclassified fundamental research. DARPA will require unlimited rights to the data generated under the resultant OT-R award.

Abstract:

For this special notice, DARPA requires that proposal abstracts be submitted before any full proposals. Please note that the deadlines for the D-PECHE study abstract and proposal submissions are different from the general abstract and proposal deadlines that are listed in HR001126S0003. **The abstract**

submission deadline for this special notice is May 27th, 2026. Abstracts must be submitted in accordance with the instructions in HR001126S0003 Attachment A, “Abstract Instructions and Template”. Information on how to submit abstracts and proposals can be found on under Section III, “Submission information”, with additional instructions for use of the Broad Agency Announcement Tool (BAAT) found here: <https://www.darpa.mil/about/offices/contracts-management/proposer-general-terms>. Proposal abstracts may not be submitted by fax or e-mail; any so sent will be disregarded.

In accordance with HR001126S0003, abstracts will be evaluated using the following criteria, listed in descending order of importance:

- (1) Overall Scientific and Technical Merit;
- (2) Potential Contribution and Relevance to the DARPA Mission;
- (3) Cost and Schedule Realism;

A detailed description of each evaluation criterion is provided in Section II of HR001126S0003.

Selected proposers will be invited to provide an OPP and participate in a virtual oral presentation to DARPA. If DARPA does not invite the proposer to submit an OPP, DARPA will provide feedback to the proposer regarding the rationale for this decision. Proposers should note that an invitation to submit an OPP is not a guarantee that an OPP based on the abstract will ultimately be selected for award negotiation. Note that proposers must submit an abstract(s) in response to this Special Notice to be considered for participation in the D-PECHE seedling. Proposers will not be invited to submit an OPP and provide an oral presentation without participating in the abstract phase of the solicitation.

Abstracts must not exceed 2 pages (excluding the title page and references). Abstracts not meeting the format described above and in HR001126S0003 will be considered non-conforming and will not be considered for review. DARPA intends to use electronic mail correspondence regarding DARPA-SN-26-67. All administrative correspondence and questions on this special notice, including requests for information on how to submit an abstract or full proposal to the BAA, should be submitted to the BAA Coordinator e-mail address: D-PECHE@darpa.mil. Questions can be submitted to the D-PECHE email inbox up until May 20th, 2026.

DARPA BTO encourages the use of the Federal Contract Opportunities website, <http://www.SAM.gov>, for retrieving HR001126S0003 and any other related information that may subsequently be provided.

Oral Proposal Package (OPP):

Specific instructions for the OPP (including content submission guidelines), in addition to oral presentation details, will be provided in the invitation letter. The information provided below is for informational purposes only. The OPP invitation will take precedence over any instructions provided below. Oral presentations are expected to take place approximately **4 weeks after notification of selection**. Oral Proposal Packages must include:

1. **Oral Presentation:** Oral Presentations are limited to no more than 20 slides in .pdf format and should be planned for 45 minutes total (30 minutes for slide presentations, and 15 minutes for a Q&A session). The Oral Presentation should include the following information:
 - a. **Technical approach**
 - b. **Personnel and facilities overview:** Provide information on key personnel dedicated to the research opportunity, including past efforts and qualifications. Provide information on facilities available to execute the proposed work.
 - c. **Milestones:** Include a slide for milestones from the **Task Description Document** below. This slide must match word for word the Milestone Description table that will be submitted as part of the **Task Description Document** and is not a summary.
2. **Task Description Document (TDD):** will be provided to performers at a later date.
3. **HR001126S0003 Attachment C:** Proposal Instructions and Volume II Template (Cost)
4. **HR001126S0003 Attachment D:** Streamlined Cost Proposal Spreadsheet
5. **HR001126S0003 Attachment G:** Accelerated Award Option Election and Attestation Form
6. **HR001126S0003 Attachment L:** Accelerated Contract Award Option – Model OT Agreement for Research

In addition to the above-required areas, the Government may request the proposer provide clarifying information in addition to the Oral Proposal Package. Submission instructions, due date for submitting the OPP, and the date and time of Oral Presentation will be provided with the invitation. Any questions asked by proposers must be submitted to D-PECHE@darpa.mil.

The government will evaluate information provided in the content submission (documentation), the Oral Presentation, and Q&A session as the basis of evaluation. Oral Presentations will be evaluated by DARPA with support from a panel composed of Government subject matter experts (SMEs).

Oral Presentations will be evaluated using the following criteria, listed in descending order of importance:

- (1) Overall Scientific and Technical Merit;
- (2) Potential Contribution and Relevance to the DARPA Mission;
- (3) Cost and Schedule Realism;

A detailed description of each evaluation criterion is provided in Section II of HR001126S0003.

After completing evaluation of Oral Presentations, DARPA will either: 1) notify proposer(s) that they have been selected for negotiation of an award; or 2) inform the proposer(s) that its proposed concept/technology/solution is not of continued interest to the Government.

DARPA anticipates multiple awards. The amount of funding made available under this special notice will depend on the quality of the submissions received and the availability of funds. The Government will only award Research OT agreements under this Special Notice and will not consider any other award options.

4. POINTS OF CONTACT

Technical Point of Contact:

Dr. Aric Lu, DARPA/BTO, D-PECHE@darpa.mil

All administrative correspondence and questions on this Special Notice, including requests for information on how to submit an abstract or OPP to the opportunity, should be submitted to the BAA Coordinator e-mail address: D-PECHE@darpa.mil. Answers to questions submitted in response to this special notice will be posted to the D-PECHE page on SAM.gov. Abstracts and/or Proposals may not be submitted via fax or e-mail – any so sent will be disregarded.